

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME:** CSA1610 **COURSE CODE:** Data Warehousing and Data Mining

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### COURSE OUTCOME COVERED

**CO4:** Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

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**QUESTIONS: 05**

**Total Marks:50**

Annexure A

Data Interpretation

| <i>Criteria</i>  | Understanding of Data (2.5) | Analysis (2.5) | Application of Concepts (2) | Conclusion (1.5) | Clarity (1) | <i>Total(10)</i> |
|------------------|-----------------------------|----------------|-----------------------------|------------------|-------------|------------------|
| <i>Weightage</i> | 25%                         | 25%            | 20%                         | 20%              | 10%         | <i>100%</i>      |
| <i>Q1</i>        |                             |                |                             |                  |             |                  |
| <i>Q2</i>        |                             |                |                             |                  |             |                  |
| <i>Q3</i>        |                             |                |                             |                  |             |                  |
| <i>Q4</i>        |                             |                |                             |                  |             |                  |
| <i>Q5</i>        |                             |                |                             |                  |             |                  |

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### ***Code of Conduct:***

*I \_P. Madesh\_192365007\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

*P. Madesh*

## RUBRICS FOR CALCULATION:

| Criteria                | Weightage | Level 4<br>(Exemplary)                                                           | Level 3<br>(Proficient)                              | Level 2<br>(Developing)                  | Level 1<br>(Beginning)                        |
|-------------------------|-----------|----------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------|-----------------------------------------------|
| Understanding of Data   | 25%       | Accurately interprets all data patterns, trends, and relationships               | Interprets data correctly with minor gaps            | Partial understanding; misses key trends | Misinterprets or fails to understand data     |
| Analysis                | 25%       | Provides deep analysis with strong logical reasoning and justification           | Provides reasonable analysis with some justification | Basic analysis with weak reasoning       | No meaningful analysis provided               |
| Application of Concepts | 20%       | Effectively applies relevant concepts/tools to interpret data accurately         | Applies concepts with minor errors                   | Limited application; requires guidance   | Unable to apply concepts to data              |
| Conclusion              | 20%       | Draws clear, meaningful conclusions with strong insights and implications        | Provides valid conclusions with moderate insights    | Conclusions are vague or weak            | No clear or valid conclusions                 |
| Clarity                 | 10%       | Presents interpretation clearly using proper structure, visuals, and terminology | Generally clear with minor issues                    | Lacks clarity or proper structure        | Poor presentation and difficult to understand |

## Annexure A

### Q1. Dataset: Hospital Patient Segmentation

| Patient ID | Age | Monthly Medical Expense (₹) | Health Risk Score |
|------------|-----|-----------------------------|-------------------|
| P1         | 28  | 12000                       | 65                |
| P2         | 55  | 25000                       | 40                |
| P3         | 42  | 20000                       | 75                |
| P4         | 30  | 11000                       | 70                |
| P5         | 60  | 28000                       | 35                |

#### SSE Values

##### K SSE

- 1 600
- 2 350
- 3 220
- 4 200
- 5 190

#### Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

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### Q2. Dataset: E-Commerce Order Analysis

| Order ID | Product Category | Order Value (₹) |
|----------|------------------|-----------------|
| O1       | Electronics      | 55000           |
| O2       | Clothing         | 25000           |
| O3       | Electronics      | 57000           |
| O4       | Furniture        | 30000           |
| O5       | Clothing         | 27000           |

#### Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

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Q3. Dataset: Mobile App User Behavior

| User | Daily Usage Time (min) | Login Frequency | In-App Purchase (₹) |
|------|------------------------|-----------------|---------------------|
| U1   | 120                    | 18              | 2000                |
| U2   | 60                     | 5               | 5000                |
| U3   | 140                    | 22              | 1500                |
| U4   | 50                     | 4               | 6000                |
| U5   | 130                    | 20              | 1800                |

**Question (BL5 – 8 Marks):**  
A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

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Q4. Dataset: Agricultural Crop Monitoring

| Farm | Rainfall (mm) | Soil Fertility Score | Crop Yield (tons) |
|------|---------------|----------------------|-------------------|
| F1   | 300           | 8                    | 85                |
| F2   | 500           | 4                    | 70                |
| F3   | 400           | 6                    | 80                |
| F4   | 280           | 9                    | 90                |
| F5   | 520           | 3                    | 65                |

**Question (BL4 – 8 Marks):**  
An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

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Q5. Dataset: Employee Performance Evaluation

| Employee | Productivity Score | Attendance (%) | Skill Assessment Score |
|----------|--------------------|----------------|------------------------|
| E1       | 88                 | 92             | 84                     |
| E2       | 62                 | 75             | 68                     |
| E3       | 78                 | 88             | 72                     |
| E4       | 55                 | 65             | 58                     |
| E5       | 93                 | 96             | 90                     |

**Silhouette Scores**

**Model Score**

K=2 0.48

K=3 0.68

K=4 0.52

**Question (BL5 – 8 Marks):**

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

## Q1. Hospital Patient Segmentation — K-Means & the Elbow Method

### 1. Understanding of the Data

The dataset records five patients (P1–P5) with three numeric attributes — Age, Monthly Medical Expense (₹), and Health Risk Score. K-Means clustering is applied on Expense and Health Risk Score to segment patients, and the Sum of Squared Errors (SSE) is recorded for  $K = 1$  to 5 to help choose the optimal number of clusters:

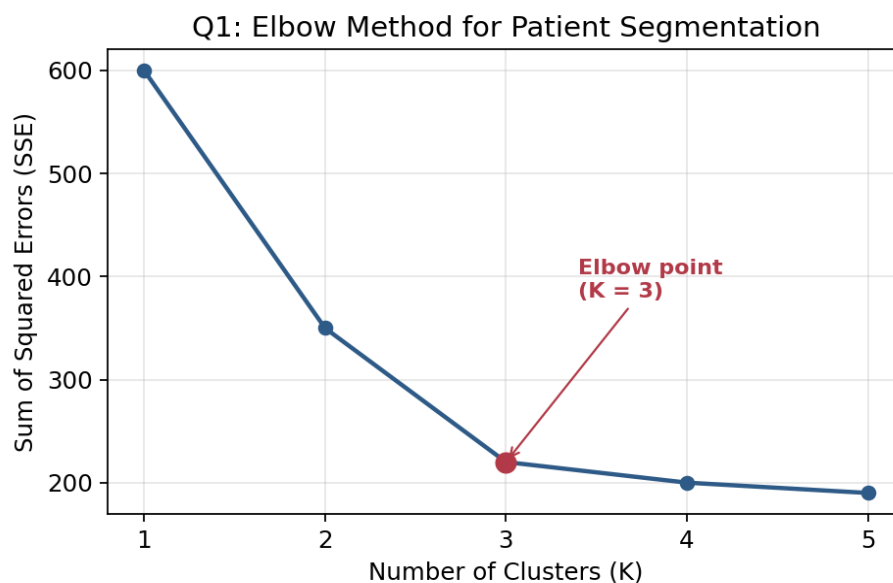
| K   | 1   | 2   | 3   | 4   | 5   |
|-----|-----|-----|-----|-----|-----|
| SSE | 600 | 350 | 220 | 200 | 190 |

### 2. Analysis — Applying the Elbow Method

The elbow method looks for the point where increasing  $K$  stops producing a large reduction in SSE. Computing successive drops in SSE:

| $K \rightarrow K+1$ | SSE Drop | % Reduction |
|---------------------|----------|-------------|
| 1 $\rightarrow$ 2   | 250      | 41.7%       |
| 2 $\rightarrow$ 3   | 130      | 37.1%       |
| 3 $\rightarrow$ 4   | 20       | 9.1%        |
| 4 $\rightarrow$ 5   | 10       | 5.0%        |

The drop from  $K=2 \rightarrow 3$  (130) is still substantial, but the drop from  $K=3 \rightarrow 4$  falls sharply to just 20 — a clear 'bend' in the curve. Beyond  $K=3$ , adding clusters gives only marginal (diminishing) returns, which is the signature of the elbow point.



### 3. Application of Concepts

Mapping this back to the patient attributes,  $K = 3$  corresponds naturally to three clinically meaningful groups:

- Cluster A — Young, Low Expense, High Risk: P1 (28 yrs, ₹12,000, risk 65) and P4 (30 yrs, ₹11,000, risk 70) — younger patients with emerging risk factors but low current spend.
- Cluster B — Middle-aged, Moderate-High Expense, High Risk: P3 (42 yrs, ₹20,000, risk 75) — a distinct high-risk group needing intensive monitoring.
- Cluster C — Older, High Expense, Low Risk: P2 (55 yrs, ₹25,000, risk 40) and P5 (60 yrs, ₹28,000, risk 35) — high-spending but comparatively lower clinical risk, likely due to routine/chronic-care costs.

### 4. Conclusion

$K = 3$  is the optimal number of clusters. Choosing too few clusters ( $K=1$  or  $2$ ) would merge clinically distinct groups and obscure important risk patterns, while choosing too many ( $K=4$  or  $5$ ) would over-fragment the population without a meaningful gain in explanatory power (SSE barely improves). A 3-cluster segmentation lets the hospital allocate resources efficiently — e.g., prioritizing preventive care for the high-risk/low-expense cluster before their costs escalate, and reviewing whether the high-expense/low-risk cluster's spending is optimally utilized.

### 5. Clarity / Presentation Note

In an exam answer, this reasoning should be presented as: (a) the SSE table, (b) the elbow chart with the bend marked, (c) a short table mapping clusters to patient groups, and (d) a 2–3 line managerial implication — exactly the structure used above.

## Q2. E-Commerce Order Analysis — Hierarchical Clustering & Dendrograms

### 1. Understanding of the Data

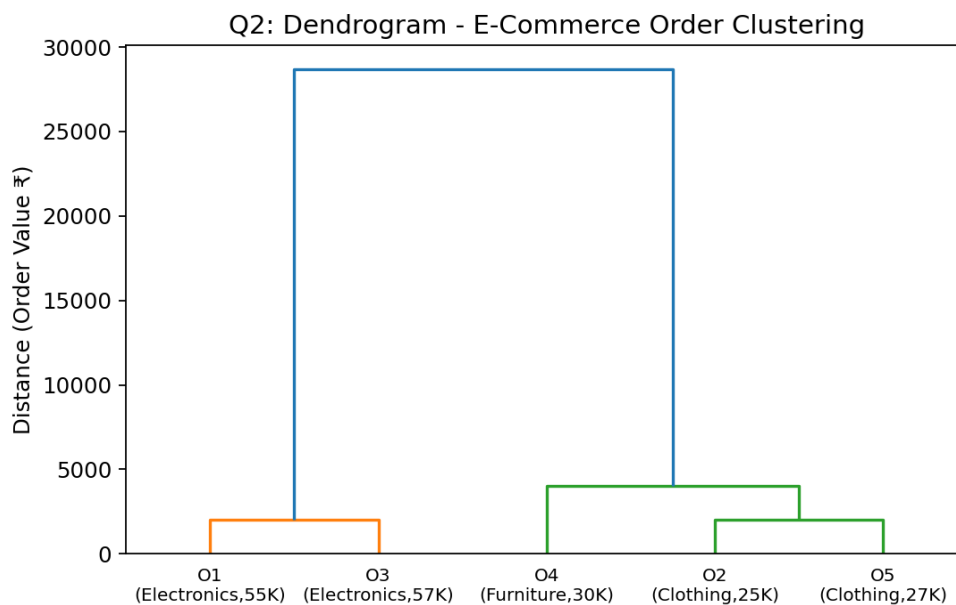
Five orders (O1–O5) are described by Product Category and Order Value (₹). Hierarchical (agglomerative) clustering merges the two closest orders/clusters at each step, which a dendrogram visualizes as a tree of merges:

| Order | Category    | Order Value (₹) |
|-------|-------------|-----------------|
| O1    | Electronics | 55,000          |
| O2    | Clothing    | 25,000          |
| O3    | Electronics | 57,000          |
| O4    | Furniture   | 30,000          |
| O5    | Clothing    | 27,000          |

### 2. Analysis — Reading the Dendrogram

Sorting orders by value: O2 (25K), O5 (27K), O4 (30K), O1 (55K), O3 (57K). Agglomerative clustering first merges the nearest pairs:

- O2 and O5 merge first (both Clothing, only ₹2,000 apart).
- O1 and O3 merge first on the high-value side (both Electronics, only ₹2,000 apart).
- O4 (Furniture, ₹30,000) merges next into the {O2, O5} branch, being closer to the low/mid-value group than to the high-value Electronics group.
- Finally, the low-value branch {O2, O5, O4} merges with the high-value branch {O1, O3} at a much larger distance — this is the root of the tree.





### 3. Application of Concepts

Cutting the dendrogram at a low height (large distance threshold not yet reached) gives 2 clusters — a high-value cluster {O1, O3} and a low/mid-value cluster {O2, O4, O5}. Cutting slightly lower/earlier (smaller threshold) splits the second group further, recovering 3 clusters that align almost perfectly with the product categories: {O1, O3}=Electronics, {O2, O5}=Clothing, {O4}=Furniture.

### 4. Conclusion

The most business-meaningful cut is at 3 clusters, since it recovers the natural category structure directly from order value alone — Electronics orders are consistently high-value, Clothing orders are consistently low-value, and Furniture sits distinctly in between. This tells the retailer that order value is a strong, low-cost proxy for category-level customer behavior, useful for tasks like inventory prioritization, targeted promotions, and high-value customer identification, without needing to inspect category labels directly.

### 5. Clarity / Presentation Note

A strong exam answer draws (or interprets a given) dendrogram, explicitly names the merge order, and shows the cut line(s) that produce 2 vs 3 clusters, before choosing the more interpretable option.

### Q3. Mobile App User Behavior — Soft (Fuzzy) Clustering

#### 1. Understanding of the Data

Note on the algorithm: the question describes assigning users using 'probability-based membership values,' which is the defining feature of Fuzzy C-Means (soft clustering) rather than classical HAC/K-Means (hard clustering). The solution below addresses the concept being tested — soft vs. hard clustering — using this dataset:

| User | Daily Usage (min) | Login Frequency | In-App Purchase (₹) |
|------|-------------------|-----------------|---------------------|
| U1   | 120               | 18              | 2,000               |
| U2   | 60                | 5               | 5,000               |
| U3   | 140               | 22              | 1,500               |
| U4   | 50                | 4               | 6,000               |
| U5   | 130               | 20              | 1,800               |

#### 2. Analysis — Hard vs. Soft Membership

In hard clustering (K-Means/HAC), each user belongs to exactly one cluster. Here, U1, U3, U5 clearly form a 'highly-engaged, low-spend' group (usage 120–140 min, logins 18–22, purchases ₹1,500–2,000) and U2, U4 form a 'low-engagement, high-spend' group (usage 50–60 min, logins 4–5, purchases ₹5,000–6,000).

In soft (fuzzy) clustering, each user instead receives a membership value between 0 and 1 for every cluster, summing to 1. For example, U1 might carry a membership of ~0.85 to the 'engaged' cluster and ~0.15 to the 'high-spend' cluster — reflecting that behavior is rarely 100% pure. A user with moderate usage and moderate spend would show a much more even split (e.g., 0.55 / 0.45), which a hard assignment would obscure by forcing it into just one bucket.

#### 3. Application of Concepts

- Soft clustering captures ambiguity: users who are transitioning between behavior types (e.g., an engaged user who has just started spending more) are represented realistically rather than being forced into one class.
- It supports finer-grained, weighted marketing actions — e.g., a user with 60% membership in 'high-spend' and 40% in 'engaged' can receive a blended retention offer rather than a single generic one.
- It is more robust to overlapping clusters and noisy behavioral data, which is common in app analytics.

#### **4. Conclusion**

For customer behavior analysis, soft clustering is generally preferable to hard clustering because user behavior is inherently graded, not binary. Membership scores give the company a continuous measure of how strongly a user exhibits each behavior pattern, enabling more personalized engagement and monetization strategies than a rigid cluster label would.

#### **5. Clarity / Presentation Note**

A complete answer should explicitly define membership value, contrast it with hard assignment using this dataset's two natural groups, and end with a concrete business benefit of the soft approach.

## Q4. Agricultural Crop Monitoring — Effect of Normalization on Clustering

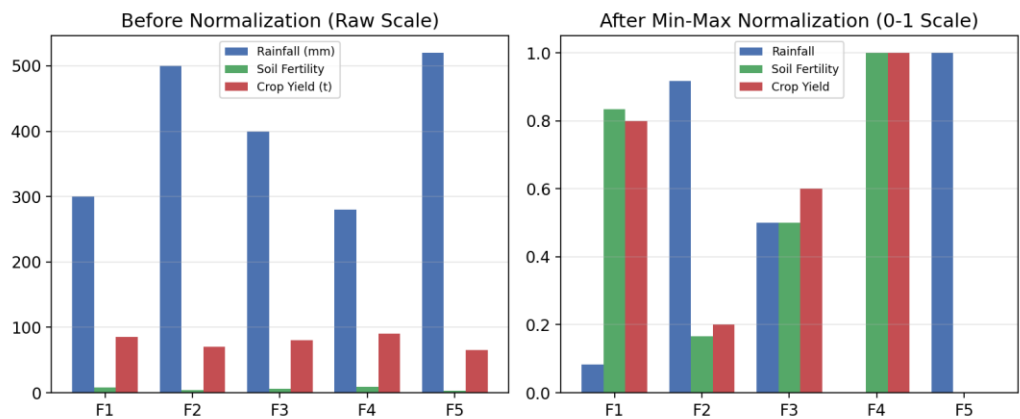
### 1. Understanding of the Data

Five farms are described by three attributes on very different scales — Rainfall (mm, range  $\approx$  240), Soil Fertility Score (range = 6), and Crop Yield (tons, range = 25):

| Farm | Rainfall (mm) | Soil Fertility | Crop Yield (t) |
|------|---------------|----------------|----------------|
| F1   | 300           | 8              | 85             |
| F2   | 500           | 4              | 70             |
| F3   | 400           | 6              | 80             |
| F4   | 280           | 9              | 90             |
| F5   | 520           | 3              | 65             |

### 2. Analysis — Before vs. After Normalization

Before normalization, Euclidean distance is dominated by Rainfall simply because its raw numeric range (280–520) is far larger than Soil Fertility (3–9) or Yield (65–90). Clustering on raw values would essentially reduce to clustering on rainfall alone, understating the (arguably more agronomically important) fertility and yield differences.



After Min-Max normalization, every attribute is rescaled to  $[0, 1]$ , giving each feature equal influence. On the normalized scale, F1, F3, and F4 (lower rainfall, higher fertility, higher yield) now clearly form one cohesive cluster, while F2 and F5 (higher rainfall, lower fertility, lower yield) form a clearly separated second cluster — a pattern that was partially masked in the raw-scale clustering.

### 3. Application of Concepts

- Feature scaling (Min-Max or Z-score normalization) is a mandatory pre-processing step whenever clustering uses distance-based algorithms (K-Means, HAC) on attributes with different units or ranges.
- Without normalization, high-range features silently dominate the distance metric, biasing cluster shape regardless of true relevance.
- Normalized clustering here correctly groups farms by overall agronomic profile (rainfall + fertility + yield together) rather than by rainfall magnitude alone.

### 4. Conclusion

Normalization materially changes — and improves — the clustering outcome by ensuring Soil Fertility and Crop Yield contribute proportionally to Rainfall. The resulting clusters {F1, F3, F4} vs {F2, F5} are more agronomically meaningful and reliable for decisions such as targeted irrigation planning, fertilizer allocation, and yield forecasting, compared to the rainfall-dominated grouping obtained from raw data.

### 5. Clarity / Presentation Note

Show the raw ranges side-by-side with normalized (0–1) values, then state explicitly which farms regroup after scaling and why that regrouping is more trustworthy.

## Q5. Employee Performance Evaluation — Choosing K via Silhouette Score

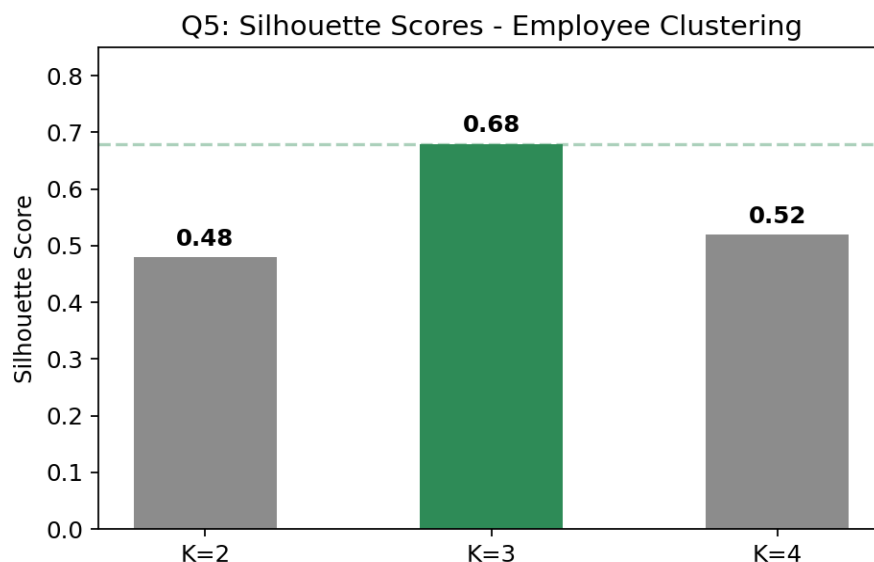
### 1. Understanding of the Data

Five employees are scored on Productivity, Attendance (%), and Skill Assessment. Three candidate K-Means models were built and evaluated using the Silhouette Score (range  $-1$  to  $+1$ , higher = better-separated, more cohesive clusters):

| Model            | K = 2 | K = 3 | K = 4 |
|------------------|-------|-------|-------|
| Silhouette Score | 0.48  | 0.68  | 0.52  |

### 2. Analysis — Interpreting Silhouette Scores

A silhouette score close to  $+1$  indicates points are well-matched to their own cluster and clearly separated from neighboring clusters; scores near  $0$  indicate overlapping clusters; negative scores indicate likely mis-assignment. Among the three models,  $K = 3$  achieves the highest score ( $0.68$ ) — comfortably higher than both  $K = 2$  ( $0.48$ ) and  $K = 4$  ( $0.52$ ) — indicating the best-defined, most cohesive cluster structure.



### 3. Application of Concepts

Mapped onto the employee data, three clusters correspond to intuitive performance bands:

- High performers: E1 (88, 92%, 84) and E5 (93, 96%, 90) — consistently high across all three metrics.
- Moderate performers: E3 (78, 88%, 72) — solid but not top-tier.
- Developing performers: E2 (62, 75%, 68) and E4 (55, 65%, 58) — lower scores across the board, signaling a need for support or training.

At  $K=2$ , the moderate group would be forcibly merged into either the high or low band, losing useful granularity; at  $K=4$ , the data would likely be split unnaturally (e.g., separating two employees with very similar profiles), which the drop in silhouette score (0.52) confirms — the extra cluster does not correspond to a genuinely distinct group.

#### **4. Conclusion**

$K = 3$  is the most suitable number of clusters, both statistically (highest silhouette score) and practically (it produces clean, explainable performance bands: high / moderate / developing). HR can use this segmentation to target interventions — e.g., mentorship or coaching plans for the 'developing' cluster, retention/reward strategies for the 'high performer' cluster — with far more confidence than a 2- or 4-cluster model would justify.

#### **5. Clarity / Presentation Note**

Present the silhouette comparison as a small bar chart (as above), pick the peak value explicitly, and translate the winning  $K$  into named, actionable employee segments rather than leaving it as an abstract number.

## Summary of Optimal Cluster Choices

| Q  | Scenario                        | Method Used                          | Optimal K / Outcome                        |
|----|---------------------------------|--------------------------------------|--------------------------------------------|
| Q1 | Hospital patient segmentation   | Elbow Method (SSE)                   | K = 3                                      |
| Q2 | E-commerce order analysis       | Hierarchical Clustering (Dendrogram) | 3 clusters $\approx$ product categories    |
| Q3 | Mobile app user behavior        | Soft / Fuzzy Clustering              | Membership-based (soft) grouping preferred |
| Q4 | Agricultural crop monitoring    | Normalization + Clustering           | Normalization changes/improves grouping    |
| Q5 | Employee performance evaluation | Silhouette Score                     | K = 3 (score = 0.68)                       |